

Computer Engineering Department

Introduction to Web Designing

Course: Website Designing

Course Code: WSD260802

Name of Staff: Mr. Pratik H. Shah

SEMESTER : I

DIVISION : A & B

INTRODUCTION TO WEB DESIGNING

I

- 1.1. Use of Internet, Terminologies used in internet, Internet verses www.
- 1.2. Web client-server computing, Client-Server Architecture, various types of server, Types of server, server side coding, client side coding,
- 1.3. Introduction to Markup languages and Scripting languages, Search Engine
- 1.4. Principles of web designing
- 1.5. Planning process of web page designing
- 1.6. Rules of web Designing
- 1.7. Introduction to web hosting

1.1

**Use of Internet, Terminologies used in internet,
Internet verses www**

Internet

The Internet is the foremost important tool and the prominent resource that is being used by almost every person across the globe.

It **connects millions of computers**, webpages, websites, and servers.

Using the internet we can **send emails, photos, videos, and messages** to anyone.

It creates a communication medium to share and get information online.

If your device is connected to the Internet then only you will be able to **access all the applications, websites, social media apps**, and many more services.

The Internet nowadays is considered the **fastest medium for sending and receiving information**.

Uses of the internet

1. Using **social media and content sharing**.
2. Instant messaging, **video conferencing, Internet Relay Chat (IRC), Internet telephony, and email** are all examples of electronic communication. These all are used through the Internet.
3. Access to **online degree programs, courses, and workshops for education and self-improvement**.
4. **Searching for jobs:** To advertise available positions, submit job applications, and hire candidates identified on social networking sites like LinkedIn, both employers and applicants use the Internet.

Other examples include:

1. Online gaming
2. Research
3. Reading electronic newspapers and magazines
4. Online shopping, or e-commerce.
5. Online discussion groups and forums

Basic Internet Terms

Domain

While every computer has its own unique address, every user using the Internet has a unique address called a **domain**. A domain recognizes one or more IP addresses. An example of a domain is weather.com and is part of the **URL** such as <https://www.weather.com>.

top-level domains are:

1. **com** - Commercial business
2. **edu** - Educational institutions
3. **gov** - Government agencies
4. **mil** - Military
5. **net** - Networks organization
6. **org** - Organizations (nonprofit)

There are additional top-level domains that are now recognized on the Internet.

1. **aero** - Air-transport industry
2. **biz** - Businesses
3. **coop** - Cooperatives
4. **info** - Unrestricted use
5. **museum** - museums
6. **pro** - Accountants, lawyers, physicians, and other professionals
7. **tv** - Television

Some countries use a sub-domain or geographical domain as part of their address. For example, an academic institution such as Oxford University in the United Kingdom can use ac.uk. An example of a URL with this domain is <http://www.ox.ac.uk/>.

Browser : A piece of software such as **Mozilla Firefox and Internet Explorer** that allows a computer to access and display documents, view pictures, hear sound, and view video clips from the World Wide Web.

E-mail: Mail that's electronically transmitted by your computer. e-mail sends your messages instantaneously, anywhere in the world. It has the capability to send messages at any time and to anyone.

File Transfer Protocol (FTP): The standard method for **downloading and uploading files over the Internet**. With FTP, you can login to a server and transfer files (meaning you can "send" or "receive" files).

Homepage: **The first page that is viewed when the browser starts.** It is also the page of a Web site that provides the introduction or content with links.

Hypertext Transfer Protocol (HTTP): The abbreviation for Hypertext Transfer Protocol. **It is the set of rules by which Web pages are transferred across the Internet.**

Internet Protocol (IP) Address: The Internet is composed of **local, regional, national, and worldwide** computer networks. Each computer on the Internet can be identified by a set of unique numbers that is called an **internet protocol (IP) address**. The IP address is composed of **four different numbers separated by periods such as 205.134.120.60**.

Link or Hypertext Link: An underlined word(s), phrase(s), or graphics on a Web page that transports the reader to additional or related information on the Internet.

TelNet: A terminal emulation protocol (or Internet program) used to **connect a computer to a remote host or server**. Telnet is one of the oldest Internet activities and is primarily used to access online databases or to read articles stored on university servers.

Uniform Resource Locator (URL): Basically the URL is the address of a computer file that has been put on a computer server to access the Internet.

Web Page: A **single hypertext file or a page** that is part of a Web site.

Website: A **collection of World Wide Web pages** or files.

Difference between Internet and WWW

S.No.	INTERNET	WWW
1	Internet is a global network of networks.	WWW stands for World wide Web .
2	Internet is a means of connecting a computer to any other computer anywhere in the world.	World Wide Web which is a collection of information which is accessed via the Internet.
3	Internet is infrastructure .	WWW is service on top of that infrastructure.
4	Internet can be viewed as a big book-store.	Web can be viewed as collection of books on that store.
5	At some advanced level, to understand we can think of the Internet as hardware .	At some advanced level, to understand we can think of the WWW as software .
6	It is originated sometimes in late 1960s .	English scientist Tim Berners-Lee invented the World Wide Web in 1989 .
7	Internet is superset of WWW.	WWW is a subset of the Internet.
8	Internet uses IP address .	WWW uses HTTP .

1.2

Web client-server computing, Client-Server Architecture, various types of server, Types of server, server side coding, client side coding

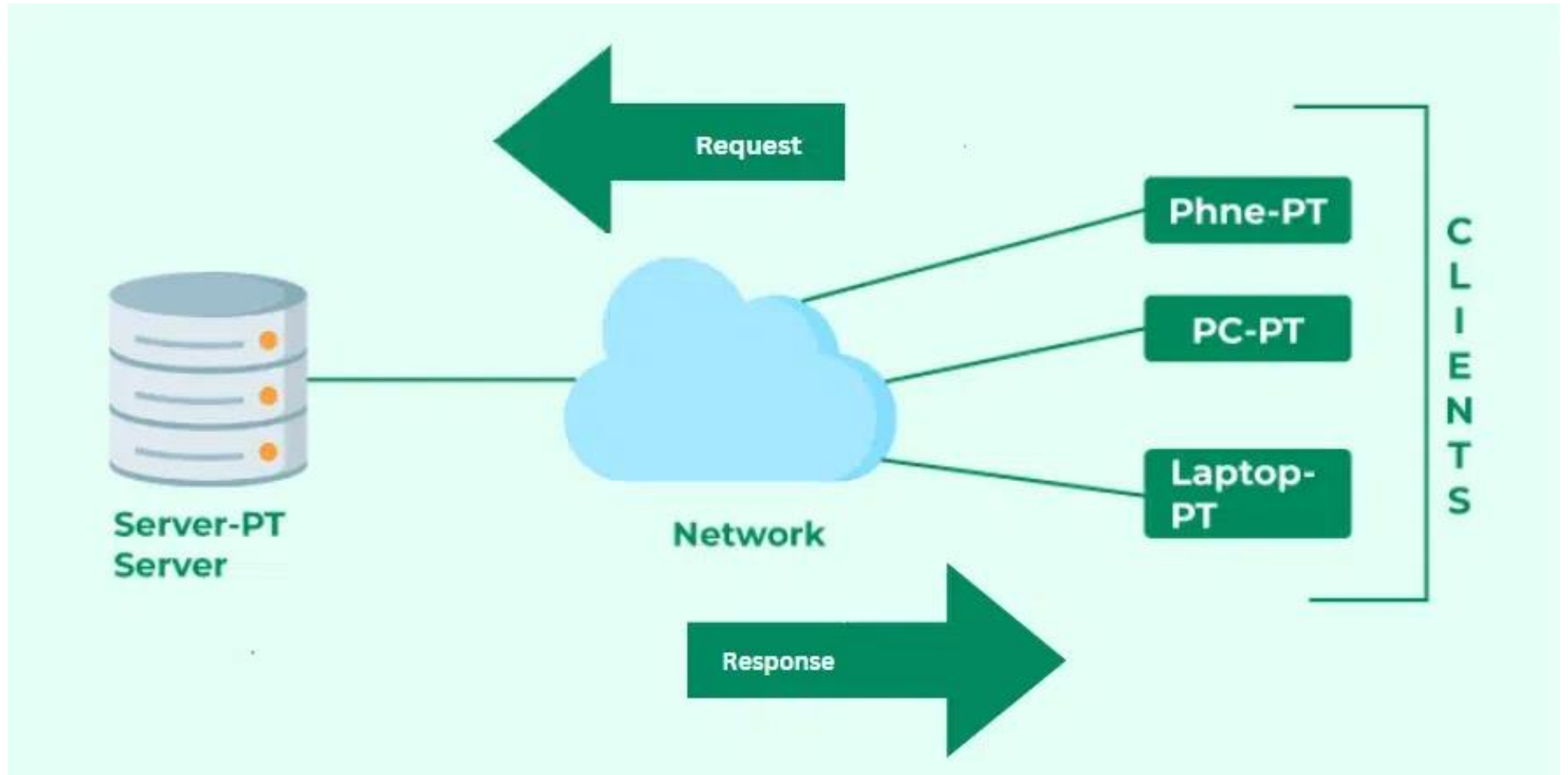
Client-Server Model

The Client-server model is a **distributed application** structure that **partitions tasks** or workloads between the providers of a resource or **service, called servers**, and service **requesters called clients**.

In the client-server architecture, when the **client computer sends a request for data to the server through the internet**, the **server accepts the requested process and delivers the data packets** requested back to the client. Clients do not share any of their resources.

Examples of the Client-Server Model are Email, World Wide Web, etc.

Client-Server Model



How Does the Client-Server Model Work?

1. **Client:** When we say the word **Client**, it means to **talk of a person or an organization** using a particular service. Similarly in the digital world, a **Client** is a computer (**Host**) i.e. capable of receiving information or using a particular service from the service providers (**Servers**).
2. **Servers:** Similarly, when we talk about the word **Servers**, It means a **person or medium that serves something**. Similarly in this digital world, a **Server** is a remote computer that provides information (data) or access to particular services.

So, it is the **Client requesting something and the Server serving it as long as it is in the database.**

Advantages of Client-Server Model

1. Centralized system with all data in a single place.
2. Cost efficient requires less maintenance cost and Data recovery is possible.

Disadvantages of Client-Server Model

1. Clients are prone to viruses, if present in the Server or uploaded into the Server.
2. Servers are prone to Denial of Service (DOS) attacks.
3. Phishing or capturing login credentials.

Types of Computer Servers

1. Web server

A web server is a computer system (either hardware or software, or both) that stores, processes, and delivers web pages and other web content to users over the internet using protocols like HTTP.

2. Proxy server

Proxy servers act as a **bridge between a host server and a client server**. A proxy sends data from a website to your computer IP address after it passes through the proxy's server. In short, A proxy server acts as an **intermediary between a user's device and the internet**, handling network requests on their behalf.

3. File transfer protocol (FTP) server

FTP servers are used to **relocate files from one computer to another**. **Uploaded files move from your computer to the server** while **downloaded files are extracted from the server onto your device**.

4. Application server

An application server is a **software framework** that provides an environment for running and managing applications, particularly those with complex business logic. It acts as an **intermediary between the user interface (front-end) and the backend systems** like databases, handling tasks such as processing user requests, managing sessions, and ensuring security.

5. File server

A file server stores data files for multiple users. They allow for faster data retrieval and saving or writing files to a computer.

6. Database server

Database servers function as large storage spaces that organizations use and access to run multiple programs to meet their needs.

7. Mail server

A mail server stores and delivers mail for clients through email service platforms.

Server-side Programming :

It is the program that **runs on server dealing with the generation of content of web page.**

- 1) Querying the database
- 2) Operations over databases
- 3) Access/Write a file on server.
- 4) Interact with other servers.
- 5) Structure web applications.
- 6) Process user input. **For example if user input is a text in search box, run a search algorithm on data stored on server and send the results.**

Examples :

The Programming languages for server-side programming are :

- 1) PHP
- 2) C++
- 3) Java and JSP
- 4) Python

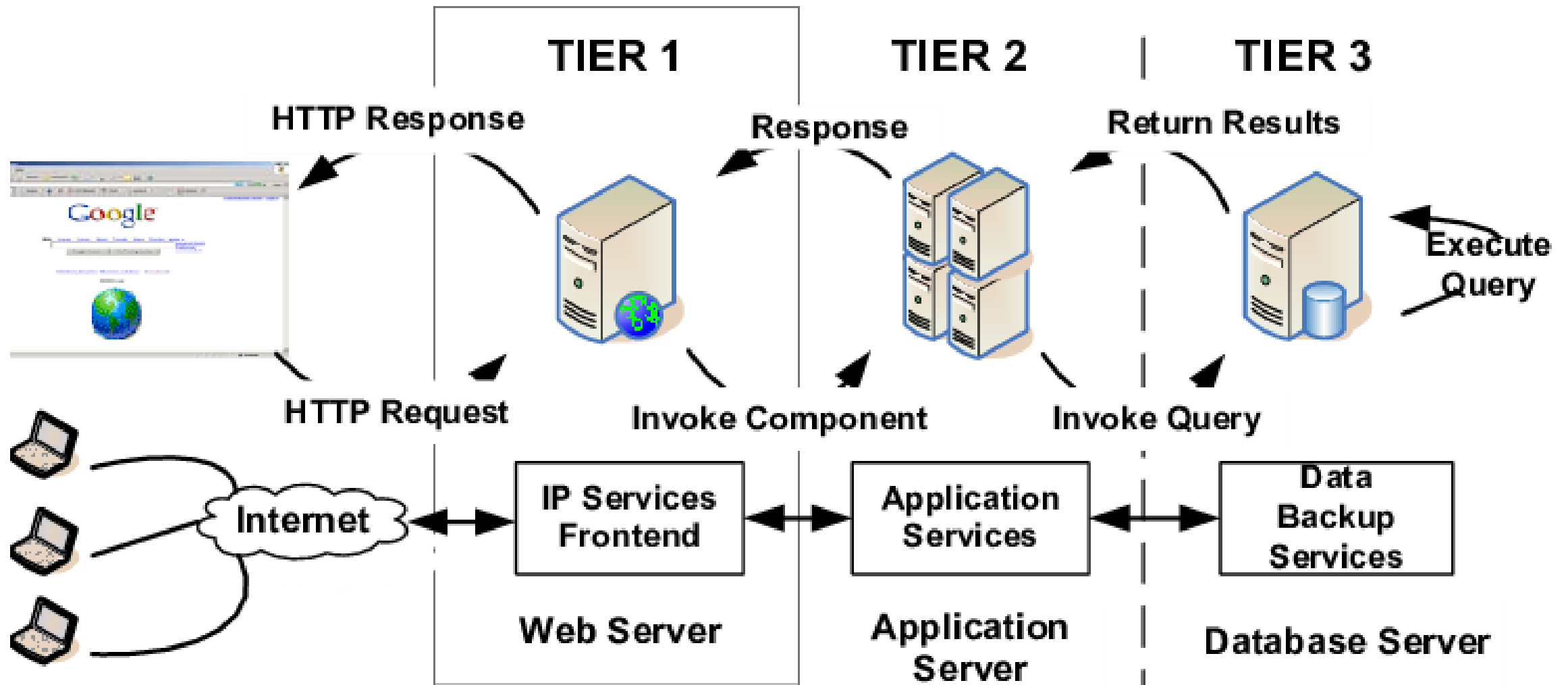
Client-side Programming :

It is the program that runs on the client machine (browser) and deals with the user interface/display and any other processing that can happen on client machine like reading/writing cookies.

- 1) Interact with temporary storage
- 2) Make interactive web pages
- 3) Interact with local storage

The Programming languages for client-side programming are :

- 1) Javascript
- 2) VBScript
- 3) HTML
- 4) CSS
- 5) AJAX



1.3

Introduction to Markup languages and Scripting languages, Search Engine

HTML, or Hyper Text Markup Language, is the standard markup language used to create web pages.

It's a combination of Hypertext, which defines the link between web pages, and Markup language, which is used to define the text document within tags to structure web pages.

HTML is human-readable and uses tags to define what manipulation has to be done on the text.

Features of HTML

1. It is easy to learn and easy to use.
2. It is platform-independent.
3. Images, videos, and audio can be added to a web page.
4. Hypertext can be added to the text.
5. It is a markup language.

Introduction to Scripting Languages

All **scripting languages** are **programming languages**. The scripting language is basically a language where **instructions are written for a run time environment**.

A scripting language is a programming language designed for integrating and communicating with other programming languages.

- **bash**: It is a scripting language to work in the Linux interface.
 - **Node js**: It is a framework to write network applications using JavaScript.
- Corporate users of Node.js** : IBM, LinkedIn, Microsoft, Netflix, PayPal, Yahoo for real-time web applications.

- **Ruby:** It is a scripting language which is great for web development.
- **Python:** It is easy, free and open source. It supports procedure-oriented programming and object-oriented programming..
- **Perl:** A scripting language with innovative features to make it different and popular. Found on all windows and Linux servers.

What are Search Engines?

Search engines are the software program that provides information according to the user query. It finds various websites or web pages that are available on the internet and gives related results according to the search.

They use algorithms to index and rank web pages based on relevance to a user's query, providing a list of results for users to explore.

Popular search engines include Google, Bing, and Yahoo.

For example: A student wants to learn C++ language so he searches the “**C++ tutorial GeeksforGeeks**” in the search engine. So the student gets a list of links that contain the tutorial links of GeeksforGeeks

Components of Search Engine

1. **Web crawler:** A search engine uses multiple web crawlers to crawl through world wide web and gather information. It is basically a software which is also known **bat or spider**.
2. **Database:** The information which is gathered by web crawler by crawling through internet is stored on the database.
3. **Search Interface:** Search interface is just an interface to the data base which is employed by the user to search through the data base.

Usage of Search Engine

Search engines have so many usages and some of them are:

1. Searching for information
2. Searching images and videos
3. Searching location
4. Searching people
5. Entertainment
6. Education

1.4

Principles of web designing

PRINCIPLES OF GOOD WEBSITE DESIGN:

Several factors such as **consistency, colors, typography, imagery, simplicity and functionality** all contribute to good website design.

1. WEBSITE PURPOSE

Your website needs to accommodate the needs of the user. Having a simple clear intention on all pages will help the user interact with what you have to offer. What is the purpose of your website? Are you imparting practical information like a 'How to guide'? Is it an entertainment website like sports coverage or are you selling a product to the user? There are many different purposes that websites may have.

2. SIMPLICITY:

Simplicity is the best way to go when considering the user experience and the usability of your website.

Color

Color has the power to communicate messages and evoke emotional responses. Finding a color palette that fits your brand will allow you to influence your customer's behaviour towards your brand. Keep the color selection limited to less than 5 colors.

Type

Typography has an important role to play on your website. It commands attention and works as the visual interpretation of the brands voice. Typefaces should be legible and only use a maximum of 3 different fonts on the website.

Imagery

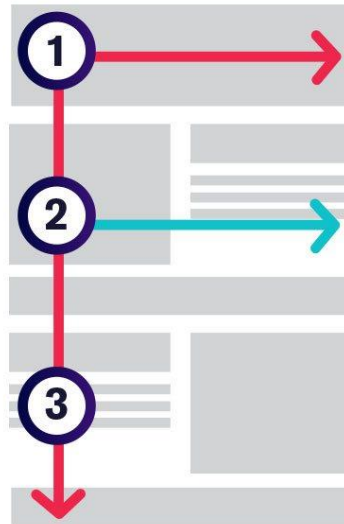
Imagery includes photography, illustration, video and all forms of graphics. Most of the initial information we consume on websites is visual and as a first impression it is important that high quality images are used to form an impression of professionalism and credibility in the visitors mind.

3. NAVIGATION

Navigation is the way finding system used on websites where visitors interact and find what they are looking for. Website navigation is key to retaining visitors. If the websites navigation is confusing visitors will give up and find what they need elsewhere. Keeping navigation simple and consistent on every page is key.

4. F-SHAPED PATTERN READING

The F- based pattern is the most common way visitors scan text on a website. Eye tracking studies have found that most of what people see is in the top and left area of the screen. The F' shaped layout mimics our natural pattern of reading in the West (left to right and top to bottom).



5. VISUAL HIERARCHY

Visual hierarchy is the arrangement of elements in order of importance. This is done either by size, color, imagery, contrast, typographically, whitespace, texture and style.

6. CONTENT

An effective web design has both great design and great content. Using compelling language great content can attract and influence visitors by converting them into customers.

7. GRID BASED LAYOUT

Grids help to structure your design and keep your content organized. The grid helps to align elements on the page and keep it clean.

8. LOAD TIME

Waiting for a website to load will lose visitors. Nearly half of web visitors expect a site to load in 2 seconds or less and they will potentially leave a site that isn't loaded within 3 seconds.

9. MOBILE FRIENDLY:

More people are using their phones or other devices to browse the web. It is important to consider building your website with a responsive layout where your website can adjust to different screens.

1.5

Planning process of web page designing

Process of website designing

There are seven steps that should be followed from start to finish when developing a website.

1. [Research and goal setting](#)
2. [Planning](#)
3. [Designing the layout](#)
4. [Writing the content](#)
5. [Coding](#)
6. [Testing and launching](#)
7. [Maintaining](#)

Website Development Planning Process:

1. RESEARCH AND GOAL SETTING

- As with any project, it is important to do proper research and set goals before beginning.
- By setting goals, it will help the website to have a direction and will also help your business to achieve specific accomplishments.
- The planning and goal setting process could take about 1-2 weeks to complete.

There are a few questions you should ask yourself during this phase:

- ✓ What do I hope for my website to accomplish?
- ✓ Who is the audience I would like to target?
- ✓ What are the main goals of the website?
- By setting goals for your website, you will be helping the site to have a clear direction and purpose.
- This can help you to determine what you want and don't want your website to have. It is also important to do some research on the target audience you hope to draw in.

2. PLANNING THE SITE:

- Planning the website involves creating a sitemap.
- This is an important step because it is kind of like the skeleton of your site.
- This process can take about 2-6 weeks to complete.
- The sitemap allows the developer to get an outline of what the site will look like, what pages there will be and how they will interact with each other. This not only helps with planning but is also beneficial to the user experience.
- A user should be able to easily navigate a site, and this begins with the development of the sitemap.
- Before you begin to plan content, a sitemap lets you design what the structure will look like.
-
- Once the sitemap is completed, the other part of this step is to create a wireframe or mock-up.
- These are just visual representations of what the site will look like.

3. DESIGNING THE LAYOUT:

- The details of the layout are what will give your website character.
- This is the step where you get to be creative with pictures, videos and what kinds of things the customer will notice when they come to your site.
- This process can take about 4-12 weeks from start to finish.
- During this step, it is especially important to keep referring back to the target audience you wish to focus on.
- Consider colors, logos, and anything that will encourage your audience to interact with the site.

4. WRITING THE CONTENT:

- This step may be going on simultaneously with the other development planning steps.
- The written content of a website is so important to its success. While this step may be happening during other steps, it is one that is crucial and deserves a lot of expertise.
- It could take from 5 to 15 weeks.
- There is a lot to consider when working on the content of the website.
- When determining what words to use, it is important that they are not too hard to understand.
- A general rule is that you have to assume not everyone is going to read words that are higher vocabulary. A website should have a vocabulary that the average person can understand.

5. CODING THE WEBSITE:

- Now that all aspects of your website have been created, you are ready to actually begin creating the website itself.
- The coding typically begins with the homepage and gradually branches out to the other pages included in the site.
- This would be where the sitemap is followed to ensure everything is coded correctly.
- The coding step could take from 6 to 15 weeks, depending on how much content and how intricate you would like your website to be.
- It is also important to set up frameworks and CMS to make sure that everything will fit onto the server during the installation process.
- Once the website is laid out according to the sitemap, it should be tested before moving any further. If all works well, then the rest of the content should be added, and formatting should be completed.

6. TESTING AND LAUNCHING:

Before the website is launched, it is crucial that it is tested out by real users.

All the links and content should be tested to see if it works.

Not only is it important to test out all the buttons and everything on the site, but it is also important to test out what users think of it.

There is user testing that can be completed to make sure the website is giving users what they need to be successful on the test.

7. MAINTENANCE:

- You might think your job is done once the website is launched, but this is not the case.
- Since technology and products are changing more rapidly than ever before, it is important to stay up-to-date with what is happening on the internet.
- Maintaining a website is hard work, but the more effort put into its maintenance, the better.
- For one, it should constantly be checked out for errors.
- When a user encounters an error, this may be frustrating and may cause them to find what they are looking for somewhere else.
- Errors can also completely block them from the information they need to make a decision on purchasing a product or service.
- This is why it is important to not only test your website for user experience before the launch, but after as well.

1.6

Rules of web Designing

10 Rules of Web Design

Keep it simple.

- Cramming too much into each page creates confusion. Visitors get frustrated when they have to scroll through a cluttered interface to find what they need.
- Keep your pages simple and your website will be easier to use.

Make a good impression.

- When you meet someone for the first time, you want to make a good first impression. The same should be true for your website.
- The overall look and feel of your site is the first thing your visitors will notice.

Provide intuitive navigation.

- Sections and pages should be well-organized with a top-down design so visitors can easily browse through the different areas of your site.

Be consistent.

- Users shouldn't feel like they are visiting a new website each time they open a new page on your site.
- A consistent layout and design across the pages within your site provides a more natural experience.

Choose your colors carefully.

- Color selection can make or break a website. When choosing colors, use a limited palette of four or five colors that don't clash.
- Make sure the text is easy to read on the background, though you can soften the contrast with colors besides black and white.

Make your website responsive.

- People access websites from a variety of devices, including smartphones, tablets, laptops, and desktop computers.
- Your website must display correctly on different screen sizes. CSS media queries are a great way to implement responsive web design.

Develop for multiple browsers.

- Test your website in multiple browsers to make sure everything appears correctly.
- It is best to catch problems ahead of time instead of relying on complaints from your visitors.

Check your website for errors.

- A small error can tarnish a great piece of work.
- If you're a webmaster, check your websites regularly for typos, broken links, and images that do not load correctly. Validate your HTML and CSS to make sure there are no syntax errors.

Write your own code.

- Whether it's HTML or PHP, nothing beats writing your code from scratch.
- If you build your site from templates and pre-written scripts, your design freedom will be limited. When you code your own pages, you have full control over how they look and act.

Don't forget the content.

- A beautiful website is an empty shell without content.
- An excellent website has both great design and great content. Make sure your pages have unique, original content that makes them worth visiting.

1.7

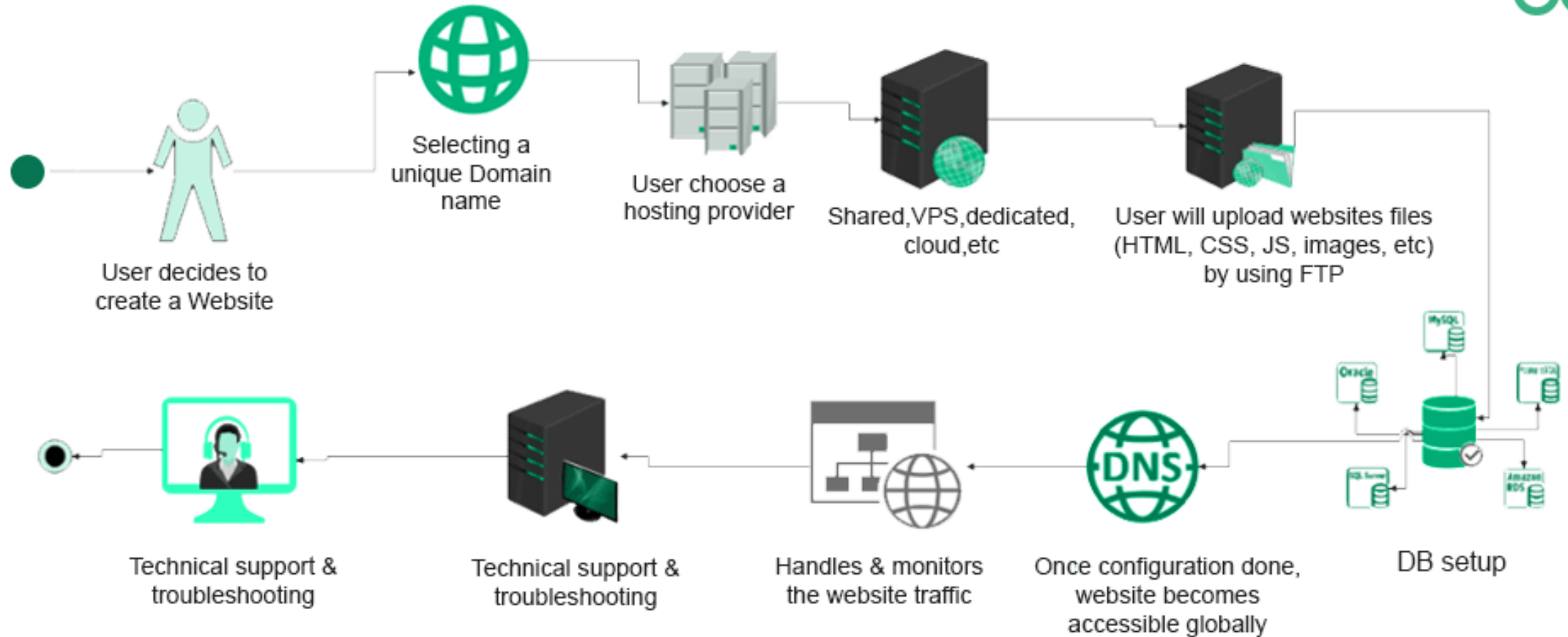
Introduction to web hosting

Web Hosting

- Web hosting is a service that allows organizations and individuals to post a website or web page onto the Internet.
- A web host, or web hosting service provider, is a business that provides the technologies and services needed for the website or webpage to be viewed in the Internet.
- Websites are hosted, or stored, on special computers called servers.
- When Internet users want to view your website, all they need to do is type your website address or domain into their browser.
- Their computer will then connect to your server and your webpages will be delivered to them through the browser.

Web Hosting is the service that provides the resources to a website and the **necessary technologies for a website that displays over the internet**. When the user hits any request the request goes to the server. Their PC connects to the server where the website is stored, then the server sends the **website's data to the user's computer** which is displayed on the screen and allows them to view it in their web browsers.

Working on Web Hosting



THANK YOU !!!!